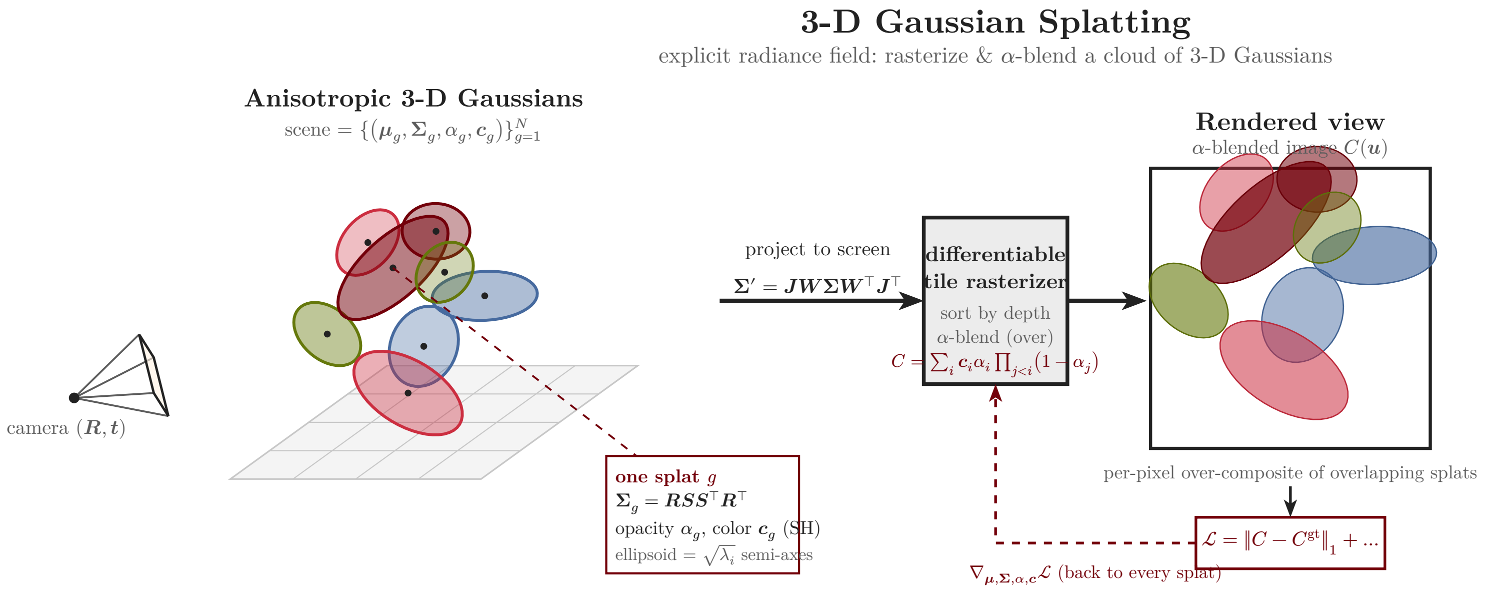

3-D Gaussian Splatting
explicit radiance field: rasterize & α-blend a cloud of 3-D Gaussians
Anisotropic 3-D Gaussians
scene $= \{(\boldsymbol{\mu}_g, \boldsymbol{\Sigma}_g, \alpha_g, \boldsymbol{c}_g)\}_{g=1}^{N}$
camera $(\boldsymbol{R}, \boldsymbol{t})$
one splat g
$\boldsymbol{\Sigma}_g = \boldsymbol{R}\boldsymbol{S}\boldsymbol{S}^\top\boldsymbol{R}^\top$
opacity α_g, color $\boldsymbol{c}_g$ (SH)
ellipsoid $= \sqrt{\lambda_i}$ semi-axes
project to screen
$\boldsymbol{\Sigma}' = \boldsymbol{J}\boldsymbol{W}\boldsymbol{\Sigma}\boldsymbol{W}^\top\boldsymbol{J}^\top$
differentiable tile rasterizer
sort by depth
α-blend (over)
$C = \sum_i \boldsymbol{c}_i\alpha_i \prod_{j<i}(1-\alpha_j)$
Rendered view
α-blended image $C(\boldsymbol{u})$
per-pixel over-composite of overlapping splats
$\mathcal{L} = \|C - C^{\mathrm{gt}}\|_1 + \ldots$
$\nabla_{\boldsymbol{\mu},\boldsymbol{\Sigma},\alpha,\boldsymbol{c}}\mathcal{L}$ (back to every splat)